

Title

Risk Stratification and Prevention Response in Complex Acute Admissions

Abstract

We performed secondary analyses on 183 patients from our prevention trial, focusing on the predictive accuracy of standardized risk assessment at different care transitions and the relationship between documented preventive actions and clinical outcomes. Formal risk assessment was documented in only 34% of prehospital encounters, 56% of emergency department admissions, and 87% of ward admissions. Among 75 patients stratified as high-risk at ward admission, 41 (55%) subsequently developed tissue breakdown, whereas among 108 low-risk patients, 15 (14%) developed injury, giving a 3.9 relative risk. Notably, tissue breakdown did not occur exclusively in high-risk patients; 23% of low-risk stratified patients experienced injury, suggesting assessment limitations. Documented preventive measures (positioning records, turning schedules) were more complete at the ward level but adherence to these protocols was inconsistent. Patients who received preventive interventions within the first 8 hours of admission had lower injury rates (16% vs 28%). The protective device was associated with 35% relative risk reduction in the subgroup receiving it within 6 hours of presentation. Cost-effectiveness analysis suggested that early systematic assessment with targeted intervention would be cost-neutral compared to treating established injuries.